EDA and ML group project NSL-KDD Dataset

Group 2
Zahra Joulaei, Will Daly, Wei Dong

Northeastern University

Problem Statement

Dataset

Network Security Laboratory – Knowledge Discovery in Databases (NSL-KDD)

The NSL-KDD dataset is a popular dataset for detecting network attacks.

- It consists of a comprehensive collection of network traffic records.
- Each record is labeled as either **normal** activity or a specific type of **attack**.

Task

The primary task involves applying machine learning techniques to detect attacks, comparing two problem formulations: binary classification for normal versus attack and multi-class classification for different attack types.

Northeastern University

Dataset & Task Overview

Why NSL-KDD?

Widely Used Benchmark

Clean Data

No Duplicates or Bias

Data Overview

==========

NSL-KDD DATASET — LOADED SUCCESSFULLY

==========

Rows : 125,973

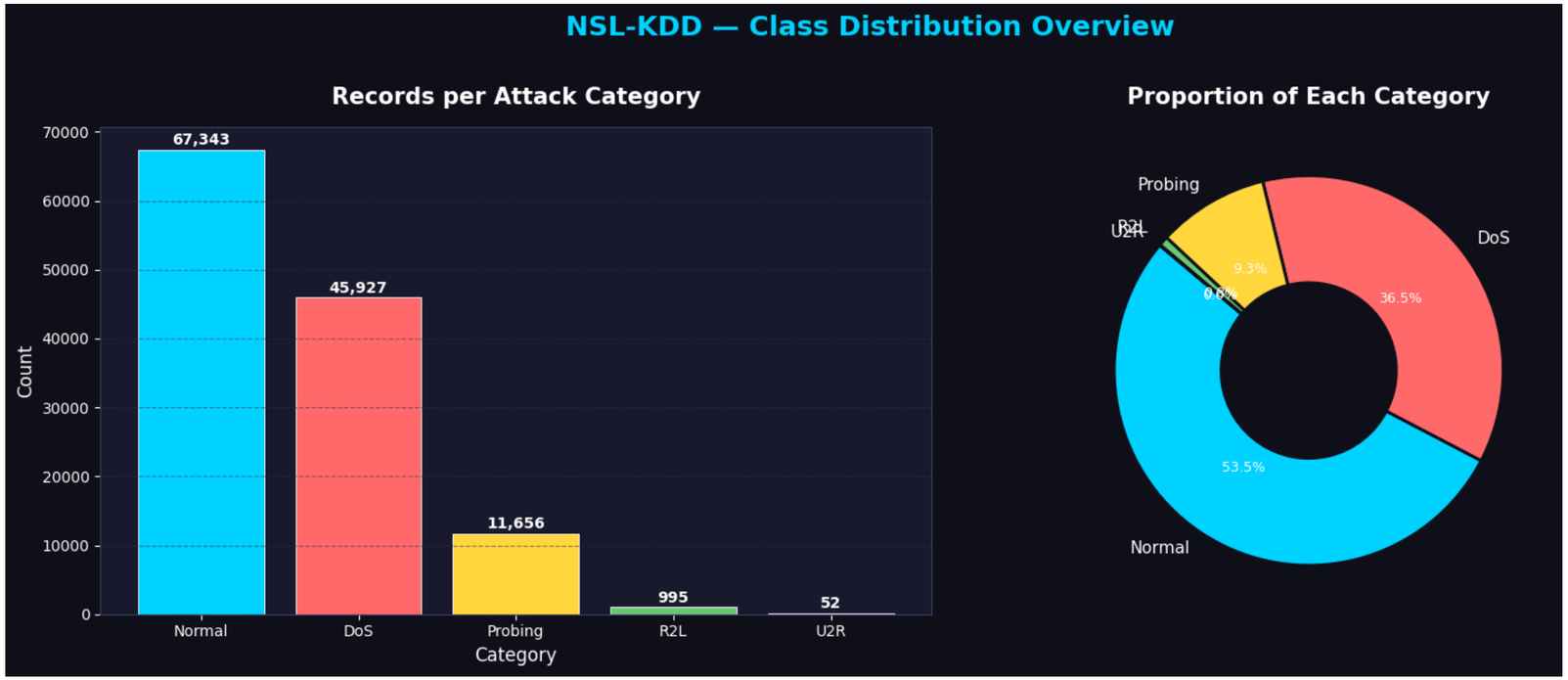
Columns : 43

Labels : 23 unique classes

==========

Northeastern University

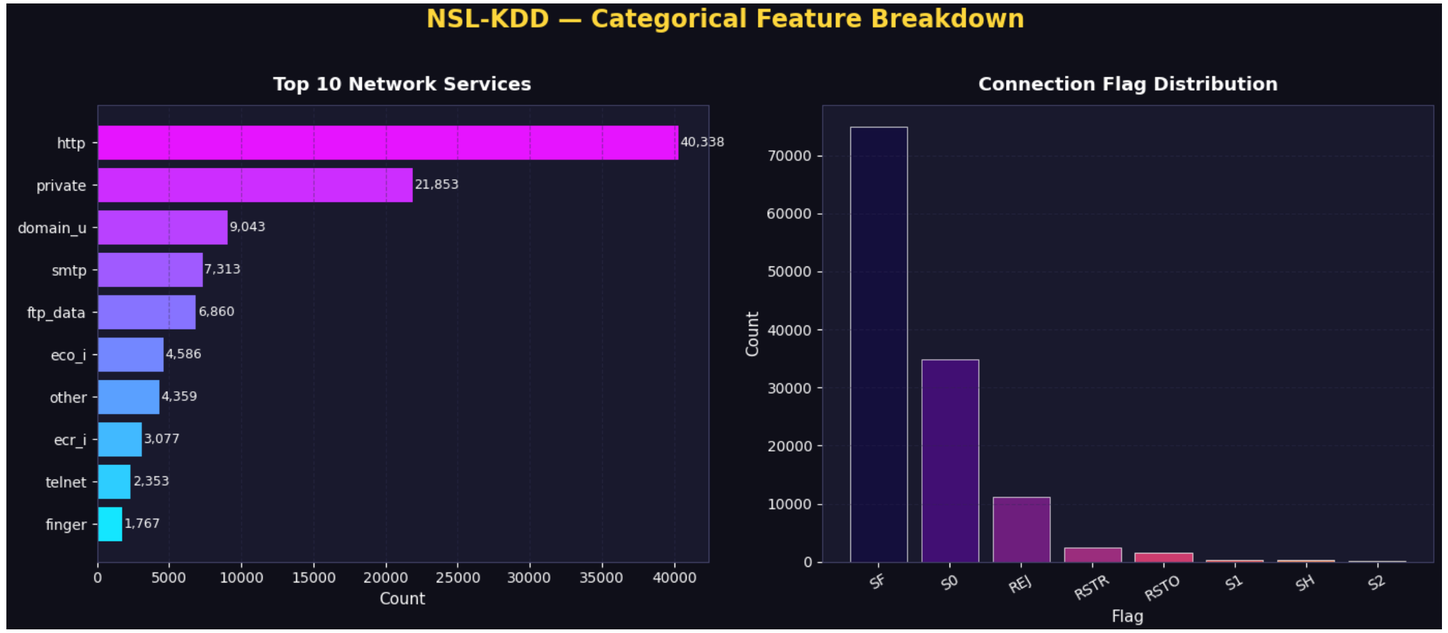
Exploratory Data Analysis



This figure shows Class Distribution overview of NSL_KDD dataset

Northeastern University

SERVICE & FLAG ANALYSIS

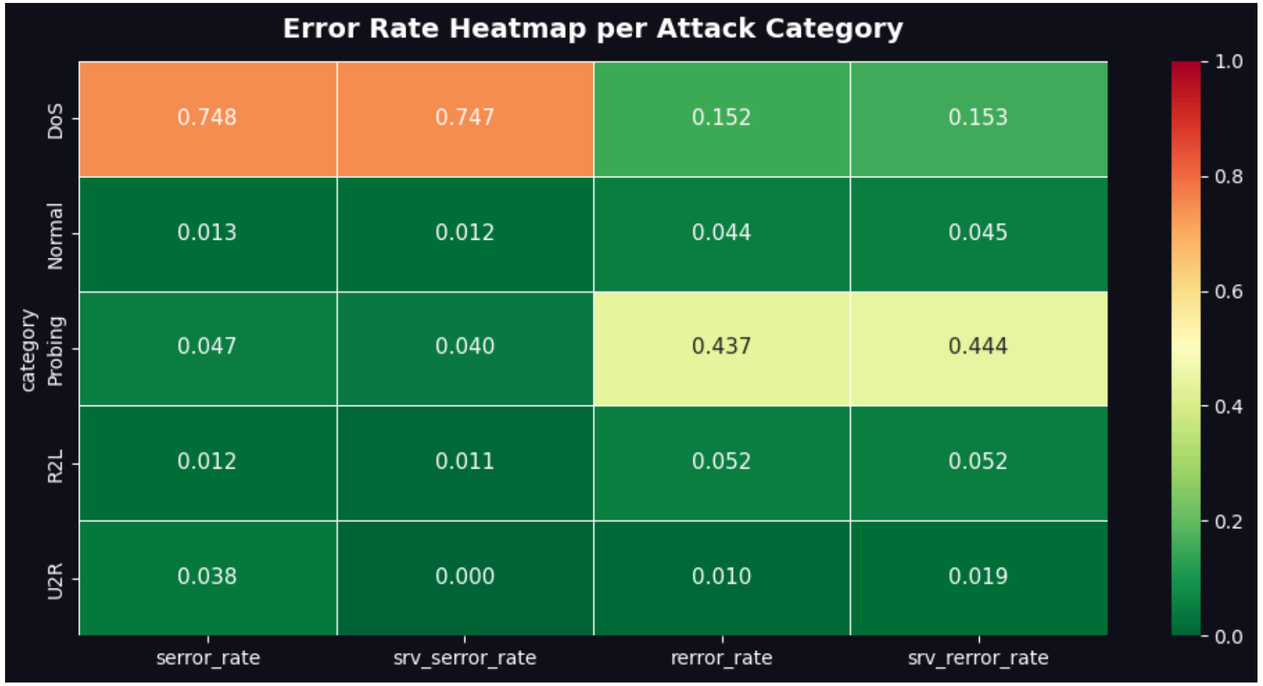



```python
# Left: top services
bars2 = axes[0].barh(top_services.index[::-1],
                     top_services.values[::-1],
                     color=sns.color_palette('cool', 10))
axes[0].set_title('Top 10 Network Services', fontsize=13,
                  fontweight='bold', color='white', pad=10)
axes[0].set_xlabel('Count', fontsize=11)
for bar, val in zip(bars2, top_services.values[::-1]):
    axes[0].text(bar.get_width() + 100, bar.get_y() + bar.get_height() / 2,
                 f'{val:,}', va='center', color='white', fontsize=9)
axes[0].xaxis.grid(True, linestyle='--', alpha=0.4)

# Right: flag distribution
axes[1].bar(flag_counts.index, flag_counts.values,
            color=sns.color_palette('magma', len(flag_counts)),
            edgecolor='white', linewidth=0.5)
axes[1].set_title('Connection Flag Distribution', fontsize=13,
                  fontweight='bold', color='white', pad=10)
axes[1].set_xlabel('Flag', fontsize=11)
axes[1].set_ylabel('Count', fontsize=11)
axes[1].tick_params(axis='x', rotation=30)
axes[1].yaxis.grid(True, linestyle='--', alpha=0.4)

plt.suptitle('NSL-KDD — Categorical Feature Breakdown',
             fontsize=17, fontweight='bold', color='#ffd93d', y=1.02)
plt.tight_layout()
plt.show()
```

SF means the connection finished normally. this is normal traffic.
SO means the connection started but never finished . a strong sign of DoS.
REJ means the connection was blocked. a sign of Probing.

Northeastern University

Error Rate Fingerprint Per Category




```python
# Calculate error rates per category
error_rate_cols = ['serror_rate', 'srv_serror_rate', 'rerror_rate', 'srv_rerror_rate']
error_table = df.groupby('category')[error_rate_cols].mean()

# Error Rate Heatmap per Attack Category
fig, ax = plt.subplots(figsize=(10, 5))
fig.patch.set_facecolor('#0f0f1a')

sns.heatmap(
    error_table,
    ax=ax,
    cmap='RdYlGn_r',
    annot=True,
    fmt='.3f',
    linewidths=0.5,
    annot_kws={'size': 11},
    vmin=0, vmax=1
)
ax.set_title('Error Rate Heatmap per Attack Category',
             fontsize=14, fontweight='bold', color='white', pad=12)
ax.tick_params(colors='white', labelsize=10)
plt.tight_layout()
plt.show()
```

SYN error = connection started but never finished. Sign of DoS attack
REJ error = connection was rejected.Sign of Probing attack

Northeastern University

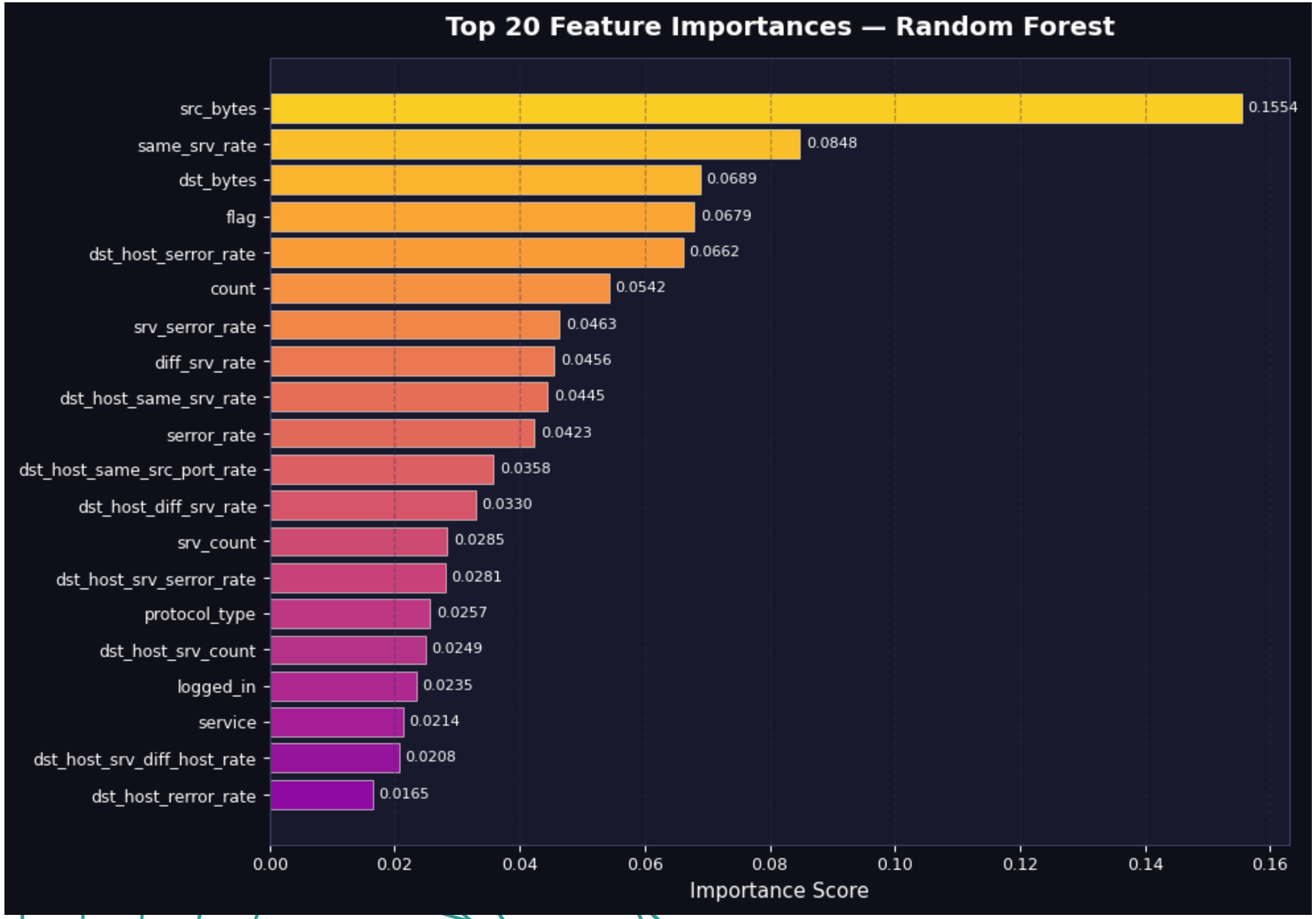
Data training

- Data Preprocessing
- Converting Categorical Features to Numerical Values
- Feature Scaling (Normalization)
- Train–Test Split (80% / 20%)
- Model Evaluation on Unseen Test Data

```
X = df_ml[feature_cols]
y = df_ml['category_enc']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

Northeastern University

RANDOM FOREST + FEATURE IMPORTANCE



Training Random Forest
Accuracy: 0.9990

Northeastern University

CLASSIFICATION REPORT

Classification Report — Random Forest

	precision	recall	f1-score	support
DoS	1.00	1.00	1.00	9186
Normal	1.00	1.00	1.00	13469
Probing	1.00	0.99	1.00	2331
R2L	1.00	0.97	0.98	199
U2R	0.89	0.80	0.84	10
accuracy			1.00	25195
macro avg	0.98	0.95	0.96	25195
weighted avg	1.00	1.00	1.00	25195

Classification Report — Logistic Regression

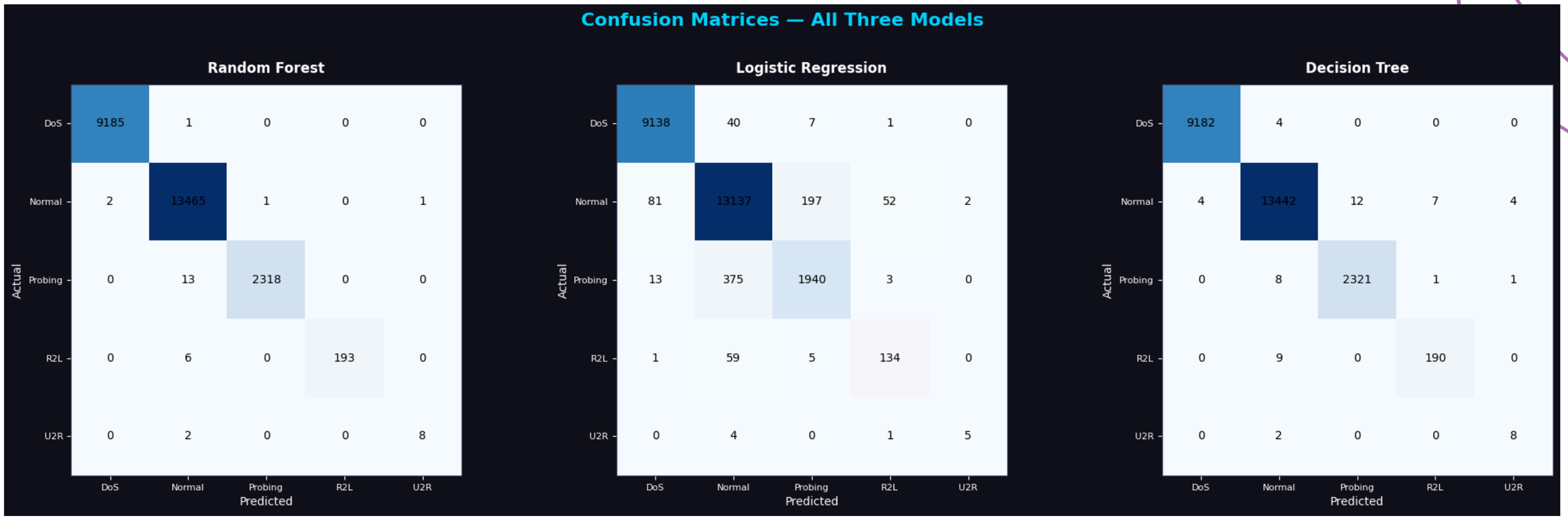
	precision	recall	f1-score	support
DoS	0.99	0.99	0.99	9186
Normal	0.96	0.98	0.97	13469
Probing	0.90	0.83	0.87	2331
R2L	0.70	0.67	0.69	199
U2R	0.71	0.50	0.59	10
accuracy			0.97	25195
macro avg	0.85	0.80	0.82	25195
weighted avg	0.97	0.97	0.97	25195

Classification Report — Decision Tree

	precision	recall	f1-score	support
DoS	1.00	1.00	1.00	9186
Normal	1.00	1.00	1.00	13469
Probing	0.99	1.00	1.00	2331
R2L	0.96	0.95	0.96	199
U2R	0.62	0.80	0.70	10
accuracy			1.00	25195
macro avg	0.91	0.95	0.93	25195
weighted avg	1.00	1.00	1.00	25195

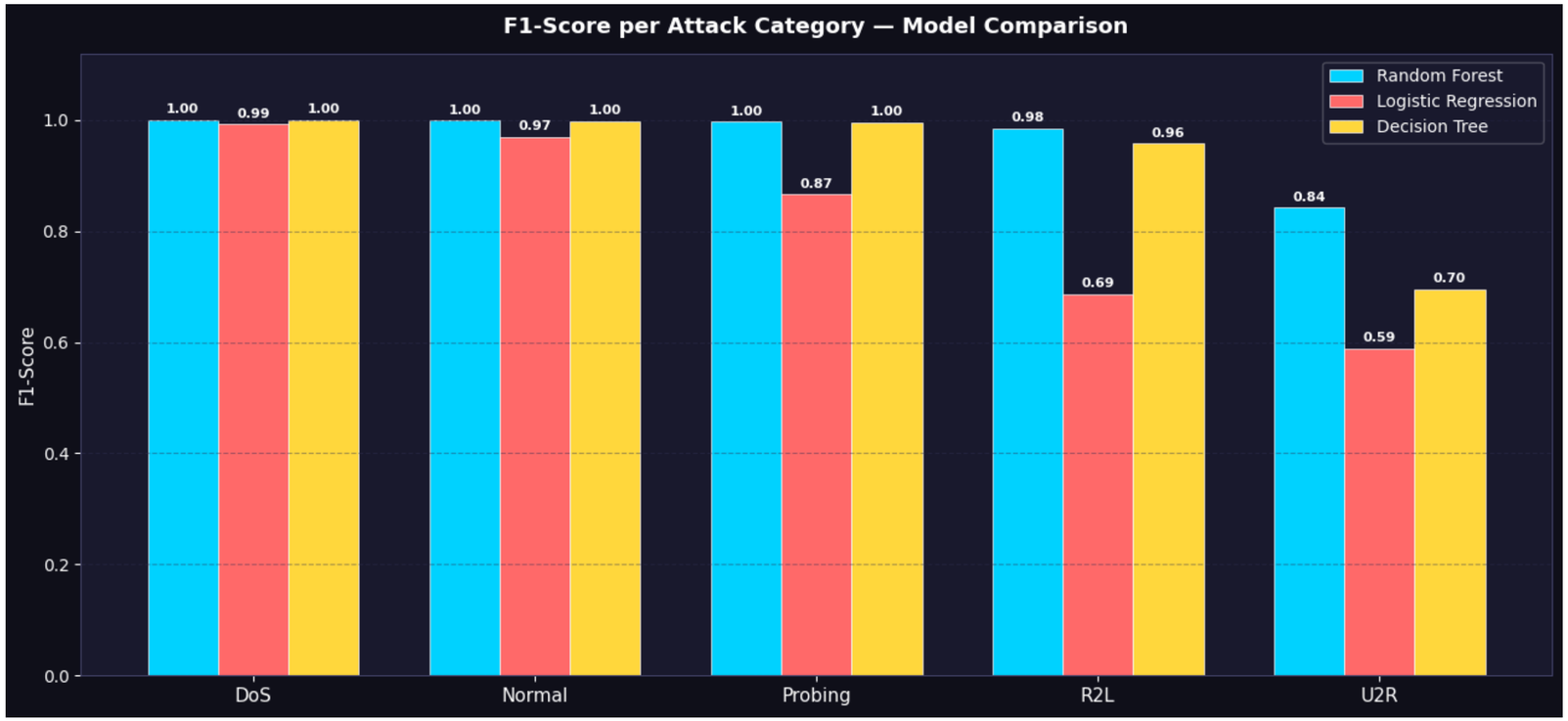
Northeastern University

CONFUSION MATRIX



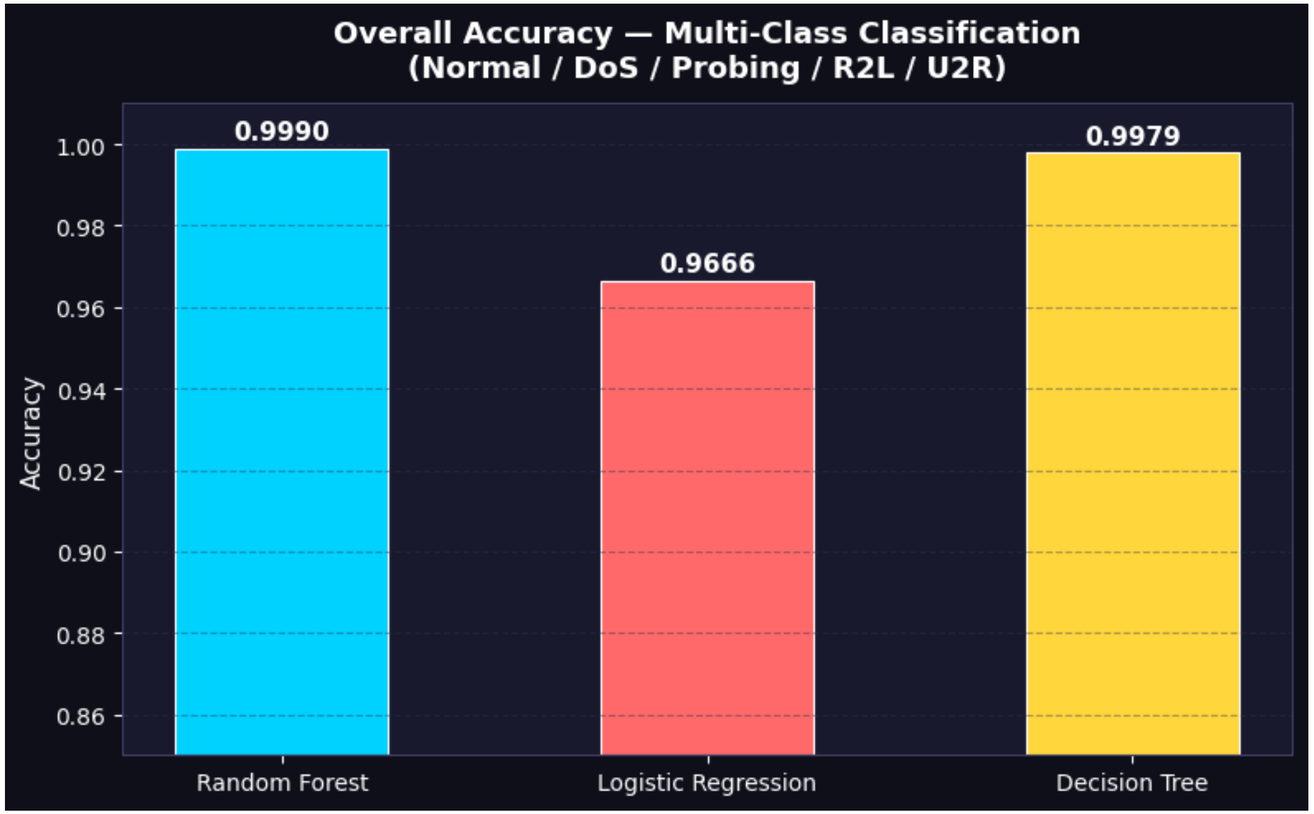
Northeastern University

F1–Score per Attack Category — Model Comparison



Northeastern University

OVERALL MODEL COMPARISON TABLE



Northeastern University

EDA and ML NSL-KDD Dataset

Group 2
Zahra Joulaei, Will Daly, Wei Dong

Thank You!